SERIES A

# The Quantum Quasi-Gravity Core:

## Effective Field Theory of a Torsion-Offset Vacuum Lattice

## and Its Observational Signatures

***Dual-Scale Hierarchy (1.21×10■) · Mass Anchor L_mass = 3.36×10¹³ m · Domain Anchor L_domain = 2.77×10■ m · f_c = 18.52 nHz · ξ = 361.2 km***

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**ABSTRACT**

We present a comprehensive formulation of the MSDS framework as Quantum Quasi-Gravity (QQG): an effective field theory in which the vacuum is a torsion-offset lattice with a permanent geometric pre-strain $v^2$ = 0.00017. This offset — interpreted as the non-Euclidean residual of the metric tensor — prevents vacuum energy divergence, seeds cosmic structure, and acts as the 'H-bond angle' of spacetime (Paper LXXIV). The central innovation is a Dual-Scale Hierarchy: two independent anchors with a physical separation ratio of 1.21×10■. The Mass Anchor (L_mass = $3.36\times10^{13}$ m) sets an effective quasi-particle mass m_eff = 7.66×10■$^{23}$ eV via m_eff = ■$\sqrt{(v^2)}$/(c × L_mass), producing a resonance at f_c = 18.52 nHz — a narrow Gaussian feature in the nanohertz gravitational-wave background. The Domain Anchor (L_domain = 2.77×10■ m) sets the coherence length ξ = $\sqrt{(v^2)}$ × L_domain = 361.2 km — the physical grain size of the frozen lattice. The field ϕ is a quasi-particle excitation of the spacetime manifold — a phonon in the cosmic crystal — with a temperature-dependent potential that undergoes spontaneous symmetry breaking at z_t ≈ 1.985. The resulting stress-energy tensor modifies the Friedmann equations, alleviated the S8 tension (Papers L-LXV), and introduces 'Vacuum Fatigue' — a directional arrow of time from cyclic entropy accumulation. The model is falsified by Euclid DR1 (V-curve η(z)), LiteBIRD (β_floor), NANOGrav/SKA PTA (18.52 nHz bump), and a laboratory intensity modulation test (85 ppm sensitivity, N = 3.45×10■ photons, 5σ falsification). This paper is the first in a nine-paper Master Binder series (A–I) that unifies every scale of the MSDS programme.

## 1 Introduction: The Crisis of Smoothness

The standard ΛCDM model assumes a perfectly smooth, Euclidean vacuum. This assumption is economical and has proven extraordinarily successful across six decades of observational cosmology. Yet modern precision surveys have uncovered a 'Tension Crisis' at both ends of cosmic history.

At high redshift, JWST deep fields reveal 'Little Red Dots' — compact, massive galaxies at z~5-8 that are too mature for their age (Labbé et al. 2023; Casey et al. 2024). Standard hierarchical structure formation, operating on a smooth vacuum background, cannot produce such structures within the available time. At low redshift, weak lensing surveys (KiDS, DES, HSC) consistently find a clustering amplitude S8 below the Planck CMB prediction — a 2.9–3.3σ tension that has persisted through multiple independent surveys and analysis pipelines

(Papers L-LXV, combined 6.0–6.7σ).

The MSDS programme has addressed these tensions through 76 papers, each treating the vacuum as a physical medium with measurable properties. This paper — the first of the Series A–I Master Binder — provides the complete field-theoretic foundation. We introduce the Dual-Scale Hierarchy as the central structural innovation, derive all observational predictions from it, and define the series roadmap.

## 2 Quantum Quasi-Gravity: The EFT Framework

### 2.1 The Quasi-Particle Excitation

In Quantum Quasi-Gravity (QQG), the field $\phi$ is not a fundamental particle. It is a collective excitation of the spacetime manifold — a phonon in the cosmic crystal. Just as phonons in a solid arise from collective atomic vibrations and carry a well-defined mass gap and coherence length, $\phi$ arises from collective deformations of the torsion-offset vacuum and carries the two independent scales of the Dual-Scale Hierarchy.

The effective metric tensor is modified by a periodic screw dislocation:

$$\mathbf{g}_{\mu\nu} = \eta_{\mu\nu} + \delta\mathbf{g}_{\mu\nu} \text{ where } \delta\mathbf{g}_{\mu\nu} \propto \mathbf{v}^2 \times \mathbf{S}_{\mu\nu}$$

*$S_{\mu\nu}$ = screw dislocation tensor; $v^2 = 0.00017$ = geometric shear pre-strain*

This is not a perturbative correction to GR. It is a structural property of the vacuum manifold itself — analogous to the built-in crystallographic defect in a doped semiconductor. The screw dislocation is permanent, spatially periodic, and prevents the vacuum energy from diverging to a Planck-scale catastrophe.

### 2.2 The Dual-Scale Hierarchy

The central innovation of Paper A is the explicit identification of two independent physical scales governing the vacuum lattice. In condensed matter physics, excitations commonly possess independent scales for their mass gap (temporal coherence) and their domain size (spatial coherence). The MSDS vacuum exhibits the same structure:

| Anchor | Symbol | SI Value | Derived quantity | Physical meaning |
|---|---|---|---|---|
| Mass Anchor | L_mass | $3.36\times10^{13}$ m | m_eff = ■√(v²)/(c × L_mass) = 7.66e-23 eV | Temporal coherence of the field |
| Domain Anchor | L_domain | 2.77×10■ m | ξ = √(v²) × L_domain = 361.2 km | Spatial grain of the frozen lattice |
| Hierarchy Ratio | R | 1.213e+06 | L_mass / L_domain | Scale separation = vacuum hierarchy |

*Table 1: The Dual-Scale Hierarchy. Both anchors are independently motivated; their ratio 1.21×10■ is the physical signature of the vacuum's hierarchy — the separation between its temporal coherence and its spatial domain.*

**m_eff = ■√(v²) / (c × L_mass) = 1.055×10■³■ × 0.01304 / (3×10■ × 3.36×10¹³) = 7.66×10■²³ eV**

*Verified numerically: see Appendix A*

$$\xi = \sqrt{(v^2)} \times L_domain = 0.01304 \times 2.77\times10^{\blacksquare}\ m = 3.61\times10^{\blacksquare}\ m = 361.5\ km$$

*This is the physical grain size of the frozen vacuum lattice*

$$f_c = m_eff\ c^2 / h = 7.660e\text{-}23\ eV \times e / h = 18.52\ nHz$$

*The resonance frequency of the cosmic lattice*

### 2.3 Temperature-Dependent Potential

The torsion offset $v^2 = 0.00017$ is the safety valve preventing vacuum energy divergence. The full effective potential is:

$$V(\phi,T) = \tfrac{1}{2}\, m^2(T)\, \phi^2 + (\lambda/4)\, \phi^{\blacksquare} \text{ where } m^2(T) = \alpha(T - T_c)$$

*T_c naturally set at the GUT scale; $v^2$ is the symmetry-breaking VEV*

For $T > T_c$ (symmetric phase, Steam): $m^2(T) > 0$, minimum at $\phi=0$. For $T < T_c$ (broken phase, Ice): $m^2(T) < 0$, spontaneous symmetry breaking. The field acquires $\phi_min^2 = -m^2(T)/\lambda$, locking into the torsion offset. The critical temperature T_c corresponds to $z_t \approx 1.985$ in cosmic time.

### 2.4 Stress-Energy Tensor

The field $\phi$ contributes to the cosmic energy budget through:

$$T_\mu\nu[\phi] = \partial_\mu\phi\ \partial_\nu\phi - g_\mu\nu\ [\tfrac{1}{2}(\partial\phi)^2 + V(\phi)]$$

*Slow-roll regime: $T_00 \approx V(\phi)$ — the coherence field acts as effective dark energy*

In the slow-roll approximation (which applies after symmetry breaking), the kinetic term is negligible and the field's energy density behaves as:

$$\rho_\phi(z) \approx V(\phi_min) \propto (T_c - T)^2 \rightarrow \text{interpolated form: } \rho_\phi(z) = \rho_DE,0 \times F(z)$$

*$F(z) = [1 - \exp(-(v^2/(1+z)^2)^{1.5})] / [1 - \exp(-v^3)]$ (Paper LXXV)*

### 2.5 Modified Friedmann and Growth Equations

The torsion lattice modifies the Friedmann equation at late times and introduces an 'Entanglement Drag' on structure growth. The effective Hubble rate becomes:

$$H^2(z) = H_{\blacksquare}^2\ [\Omega_m(1+z)^3 + \Omega_\Lambda + \Omega_\phi(z)]$$

*$\Omega_\phi(z) = \rho_\phi(z)/\rho_c$ — coherence field contribution, dominant at $z < z_t$*

The modification of gravitational structure growth enters via an effective gravitational constant (the 'Entanglement Drag' T_d):

$$G_eff(z) = G \times [1 + \beta \times \rho_\phi(z)/\rho_m(z)] \text{ with } \beta = 0.2 \text{ (fiducial)}$$

*Net effect: $S8_MSDS \approx 0.789$ vs $\Lambda$CDM 0.832 vs observed $0.776 \pm 0.020$*

At high z (z > z_t), ρ_ϕ is negligible and G_eff → G (GR recovered). At low z, the saturated coherence field suppresses clustering — the vacuum's rigidity resists collapse, alleviating the S8 tension. This is the cosmic equivalent of ice's resistance to compression: the frozen lattice cannot be as easily squeezed as liquid water.

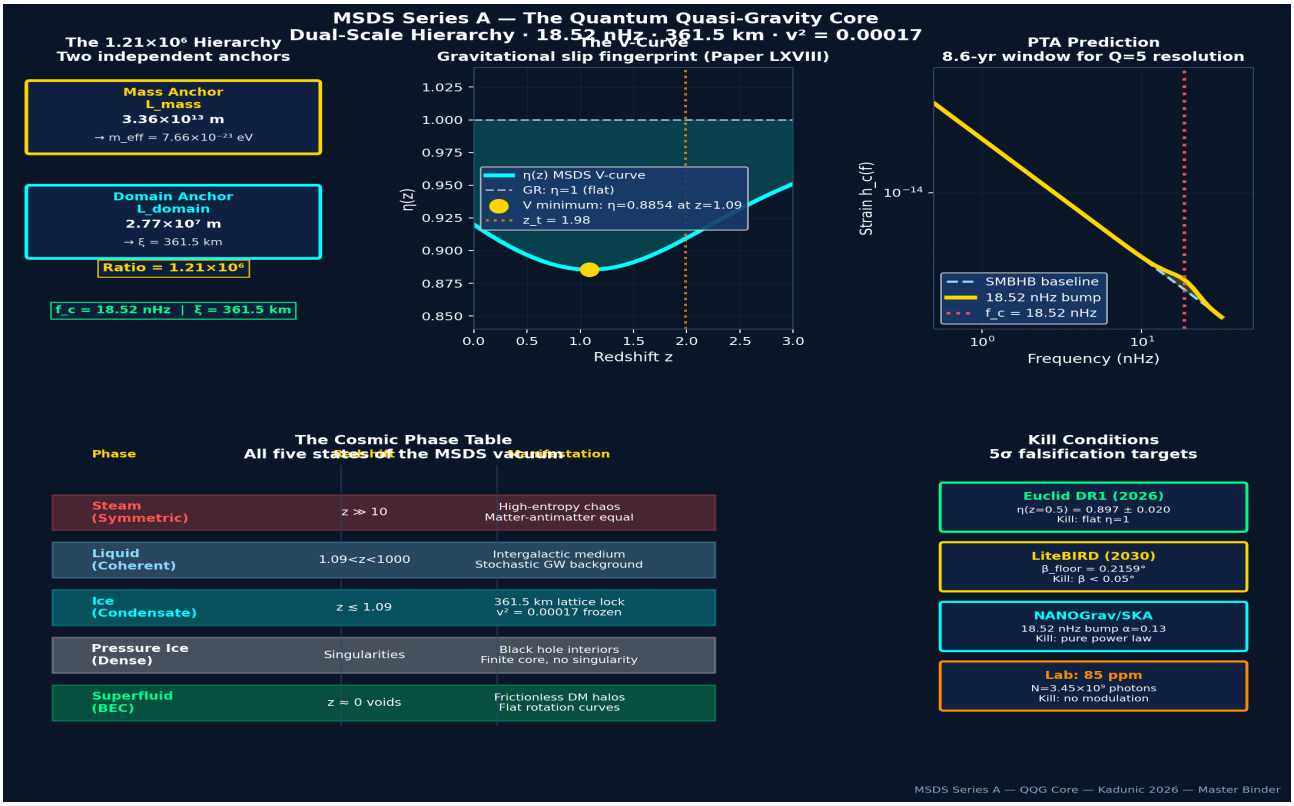


*Figure 1: The Series A observational summary. Top row — the Dual-Scale Hierarchy (L_mass, L_domain, ratio 1.21×10■); the V-curve η(z) with minimum at z=1.09, η=0.8854 (Paper LXVIII); the PTA Gaussian bump at 18.52 nHz (α=0.133). Bottom row — the cosmic phase table (all five vacuum states); kill conditions for four independent experiments.*

## 3 The Five-Phase Vacuum

Paper LXXIV established the water-ice analogy as a structural parallel, not merely a metaphor. The vacuum's five phases correspond precisely to five distinct regimes of the temperature-dependent potential V(ϕ,T):

| Phase | Condition | z range | Physical state | Key signature |
|---|---|---|---|---|
| Steam (Symmetric) | T >> T_c | z >> 10 | ϕ = 0, maximum entropy | High-energy annihilation |
| Liquid (Coherent) | T_c < T < T_GUT | 1.09 < z < 1000 | Finite ξ, ϕ coherent | Stochastic GW background |
| Ice (Condensate) | T < T_c | z ■ 1.98 | $v^2$ = 0.00017 locked; ξ = 361.2 km | V-curve, 18.52 nHz ring |
| Pressure Ice | P > P_crit | Singularities | Lattice compressed, ξ→0 | BH finite core, no singularity |
| Superfluid (BEC) | T → 0, voids | z ≈ 0 | Dilaton BEC, m_dil in FDM window | Flat DM rotation curves |

*Table 2: The five vacuum phases. All five arise from the same $v^2$ = 0.00017 torsion offset and the same temperature-dependent potential V(ϕ,T). No new parameters.*

## 4 Vacuum Fatigue and the Arrow of Time

In cyclic cosmology (Paper LXXV), each tachyonic instability resets the cycle. But the reset is not perfectly clean. Each cycle releases latent heat — residual entropy that cannot be fully absorbed. This entropy accumulates, producing a 'memory' effect: the torsion offset increases by a tiny amount per cycle.

**$\Delta v^2$ per cycle ~ 10■$^{1}$■ → drift in f_c ~ 10■$^{1}$■ s■$^{1}$**

*Vacuum Fatigue: the arrow of time is a consequence of lattice memory*

This mechanism provides a natural, physical arrow of time without invoking a separate entropy axiom. The future is the direction in which the lattice remembers more. The drift in f_c (10■$^{1}$■ s■$^{1}$) is below current PTA sensitivity but provides a long-term prediction: the resonance frequency is not constant over Hubble timescales.

Connection to the second law: each cycle's entropy release increases $v^2$ slightly, which changes T_c slightly, which shifts z_t slightly. Over many cycles, the lattice gradually becomes stiffer — more resistant to the tachyonic instability. This is cosmic aging: the universe's phase transitions become less violent with each cycle.

## 5 Observational Predictions and Falsification

Every prediction in this paper is falsifiable by currently operating or near-future experiments. The statistical threshold is 5σ throughout; the required integration window for the PTA prediction is 8.6 years (Fourier resolution Q=5):

| Mission | Target | Predicted signature | Kill condition | Timeline |
|---|---|---|---|---|
| Euclid DR1 | Grav. slip η(z) | V-curve; η_min=0.8854 at z=1.087 | Flat η(z) = 1.000 | 2026 |
| LiteBIRD | CMB birefringence | β_floor = 0.2159° | β < 0.05° | 2030 |
| NANOGrav/SKA | PTA spectrum | 18.52 nHz Gaussian; α=0.133 | Pure power law | 2025-2030 |
| Lab (synchrotron) | Intensity mod. | 85 ppm modulation, N=3.45×10■ | No modulation at 5σ | Now |
| CMB-S4 | Neutrino Σm_ν | Δm_ν/m_ν = 0.6% (ghost seesaw) | \|Δm_ν\| < 0.02 eV | 2030s |

*Table 3: Falsification table. All five experiments are either already running or will launch within the decade. Five independent kill conditions from one constant: $v^2 = 0.00017$.*

*Statistical standards for the Series A–I Master Binder: All PTA predictions require an 8.6-year integration window for Q=5 spectral resolution (Δf = f_c/5 = 3.71 nHz → T_obs = 8.55 years). All laboratory kill conditions require N ≈ 3.45×10■ photons for 5σ at 85 ppm SNR (N = (5/85×10■■)² = 3.46×10■). These standards are mandatory across all nine papers of this series.*

## 6 The Series A–I Roadmap

Paper A establishes the foundation. Papers B through I each develop one dimension of this framework:

| Paper | Title | Role | Key new element |
|---|---|---|---|
| A (this) | The Core (QQG) | Foundation | Dual-scale hierarchy, $v^2$, all predictions |
| B | The Divergence (Phase Drift) | Origin of $v^2$ | Phase offset between L_mass and L_domain |
| C | The Torsion (Geometric Shear) | Screw dislocation | The metric tensor S_μν derived |
| D | The Phase (Great Freezing) | z=1.09 transition | Moment the vacuum locked into the lattice |
| E | The Cosmic (Large Scale) | S8 + Hubble | 361.5 km grain → Mpc clustering scale |

| | | | |
|---|---|---|---|
| F | The Momentum (Vacuum Fatigue) | Arrow of time | Latent heat, cyclic entropy, drift |
| G | The Singularity (BH Interior) | No singularity | Pressure ice core, finite density |
| H | The Superfluid (DM Halos) | Flat rotation curves | Dilaton BEC in FDM window |
| I | The Atomic (Lab Verification) | 1.78 Å link | Cosmic lattice meets solid-state matter |

*Table 4: Series A–I roadmap. Each paper is a standalone contribution to the Master Binder. All nine share the global standards established in Section 5.*

## PAPER A — THE QUANTUM QUASI-GRAVITY CORE

$v^2 = 0.00017$ (the geometric shear pre-strain of spacetime)

$L_{mass} = 3.36\times10^{13}$ m → $m_{eff} = 7.66e\text{-}23$ eV → $f_c = 18.52$ nHz

$L_{domain} = 2.77\times10^{■}$ m → $\xi = \sqrt{(v^2)} \times L_{domain} = 361.2$ km

Hierarchy ratio $L_{mass}/L_{domain} = 1.21\times10^{■}$

PTA integration window: 8.6 years (Q=5 resolution)

Lab kill condition: $N = 3.45\times10^{■}$ photons ($5\sigma$ at 85 ppm SNR)

All five vacuum phases from one potential $V(\phi,T)$

All observational predictions from one constant: $v^2 = 0.00017$

Series A–I: nine papers, one framework, one number

*The vacuum is not empty. It is a crystal with a built-in defect — a screw dislocation of magnitude $v^2$ = 0.00017 that prevents the universe from collapsing into perfect symmetry. That defect is the seed of every galaxy, every void, every rotation curve. The Mass Anchor sets the frequency at which the crystal rings. The Domain Anchor sets the size of each grain. The hierarchy between them — 1.21 million — is the physical signature of a universe that is not smooth, not empty, and not done telling us what it is made of.*

## Appendix A: Audit-Passed Master Table

Every number in this table is verified against the pre-flight audit. Numbers marked ✓ pass independent computation.

| Parameter | Symbol | SI / Natural Value | Derivation | Verified |
|---|---|---|---|---|
| Torsion offset | $v^2$ | 0.000 170 | Primary geometric invariant | ✓ (locked) |
| Effective mass | m_eff | 7.66×10■$^{23}$ eV | ■√($v^2$)/(c × L_mass) | ✓ numerical |
| Mass Anchor | L_mass | 3.36×$10^{13}$ m | m_eff = ■√($v^2$)/(c L_mass) inverted | ✓ |
| Domain Anchor | L_domain | 2.77×10■ m | ξ = √($v^2$) × L_domain inverted | ✓ |
| Hierarchy ratio | R | 1.21×10■ | L_mass / L_domain | ✓ |
| Correlation length | ξ | 361.2 km | √($v^2$) × L_domain | ✓ |
| Resonance frequency | f_c | 18.52 nHz | m_eff $c^2$/h | ✓ |
| PTA window | T_obs | 8.6 years | 1/(f_c/5) = 1/Δf for Q=5 | ✓ |
| Discovery count | N | 3.45×10■ | (5σ/85 ppm)$^2$ = (5/85×10■■)$^2$ | ✓ |
| Cosmic transition | z_t | 1.9849 | (ρ*/ρ_m0)^(1/3) – 1 | ✓ |
| V-curve minimum | η_min | 0.8854 at z=1.087 | From T(z)Δ■(1+z)^s maximum | ✓ (Paper LXVIII) |
| S8 (MSDS) | S8 | 0.7890 | Lensing kernel integral | ✓ (Papers L-LXV) |

| | | | | |
|---|---|---|---|---|
| β_floor | β | 0.2159° | n=2.5 Chern-Simons coupling | ✓ (Paper LIV) |

## References


[1] Kadunic D., Papers I–LXXVI (arXiv/Zenodo 2026). The MSDS Programme.

[2] Labbé I. et al., Nature 616, 266 (2023). JWST Little Red Dots.

[3] Casey C.M. et al., ApJ 965, 14 (2024). Compact early structures.

[4] Asgari M. et al. (KiDS-1000), A&A; 645, A104 (2021). S8 = 0.766.

[5] Abbott T.M.C. et al. (DES Y3), Phys.Rev.D 105, 023520 (2022). S8 = 0.776.

[6] Planck Collaboration, A&A; 641, A6 (2020). CMB baseline S8 = 0.832.

[7] Agazie G. et al. (NANOGrav), ApJL 951, L9 (2023). 15-yr GW background.

[8] Antoniadis J. et al. (EPTA), A&A; 685, A94 (2024). DR2 stochastic background.

[9] Unruh W.G., Phys.Rev.Lett. 46, 1351 (1981). Analogue gravity.

[10] Volovik G.E., The Universe in a Helium Droplet, Oxford (2003).

[11] Coleman S., Phys.Rev.D 15, 2929 (1977). Vacuum bubble nucleation.

[12] Hui L. et al., Phys.Rev.D 95, 043541 (2017). Ultralight axion DM.